KA SHING BILL, KU

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EDUCATION

University of Oxford | MEng Master of Engineering **Oct 2022 – July 2026**
Exeter College
Grade: 2:1 | Master's Thesis grade: 78%
Master's Thesis topic: Towards more human-like learning for artificial neural network language models
Supervisor: Prof Janet B. Pierrehumbert
Master's modules:

- C18 Machine Vision and Robotics
- C19 Deep Learning
- C20 Robust and Distributed Control
- C21 Optimal and Learning-Based Control
- C24 Probability, Systems and Perturbation Methods
- C29 Computing Technology

EXPERIENCES

Master's Thesis, University of Oxford | Grade: 78% (Distinction) **2025 - 2026**
Towards more human-like learning for artificial neural network language models
Supervisor: Prof Janet Pierrehumbert

- Designed and executed a systematic, multi-condition study of phonemic and prosodic text representations, extending prior BabyLM research (Goriely et al. 2024) with a seven-condition information ladder across 10M and 100M token corpus scales
- Trained 14 BERT-family masked language models (BERT-Small and BERT-Base) from scratch, implementing the full pipeline from raw corpus preprocessing and BPE tokenizer training to masked language modelling and evaluation
- Engineered a scalable NLP preprocessing pipeline in Python integrating phonemic transcription (CMU Pronouncing Dictionary, eSpeak), rule-based and BERT-based prosodic boundary annotation (ToBI framework, spaCy), and custom BPE tokenisation, applied to the BabyLM corpus
- Designed and implemented two purpose-built evaluation suites: a phonotactically-controlled lexical decision task with PLL and frozen-CLS classifier probes, and a three-part semantic benchmark covering concept-boundary discrimination, semantic category induction (k-means and agglomerative clustering, ARI/NMI), and graded familiarity scoring
- Utilized the Oxford Advanced Research Computing (ARC) cluster with SLURM job scheduling, built a self-contained software environment on the cluster with Linux CLI through the terminal. Trained the language models across 4× NVIDIA H100 GPUs using bfloat16 mixed precision and distributed data loading, reduced the training time of 92M BERT-base class model to ~5hrs
- Demonstrated that prosodic annotations model-based prosodic annotation yields the strongest semantic category representations across independent evaluation tasks, providing empirical support for the prosodic bootstrapping hypothesis in neural language models

AI Safety Collab Summer 2025, The European Network for AI Safety **2025**
AI Alignment track by AI Safety Altas

- Conducted research on topics such as AI alignment, risk mitigation strategies, governance, interpretability, and evaluation of advanced AI systems
- Presented weekly findings and contributed to group discussions on AI safety strategies and ethical deployment
- Collaborated with peers to explore methods for ensuring AI systems remain safe, align with human values and beneficial to society

Model Difference Amplification (GitHub Repo here) | Personal research **2025**

- Conducted research on the method of model difference amplification in surfacing rare, undesired behaviours in LLMs during fine-tuning from corrupted datasets
- Based on the Microsoft Phi-3-4k-instruct model, a 'corrupted' model is created by fine tuning it with the MedQuad dataset, with 5% of the data intentionally augmented with unprofessional terms.
- Evaluated the coherence and degradation of the LLM responses as a function of alpha, the amplification factor.
- The coherence score of each model response to an independent input dataset is rated by Claude 3.5 Haiku via Anthropic API, with the results plotted against different amplification factors.

3rd Year Research Project, University of Oxford **2024 - 2025**
Designing Intelligent Music Technology: Smart Backing Tracks
Supervisor: **Prof David De Roure** and **Dr Kevin Page**

- An industry-first attempt to unlock real time spontaneous backing track adjustments in live performances through natural language inputs from the Music Director, with the whole system running on-device.
- Trained a two-classifier system for Text-to-Intent classification, consisting of a fine-tuned BERT and a Multilayer Perception (MLP) neural network.
- Context-aware by retrieving real time musical context from the DAW
- Adapts to each user's unique style and phrase preference when speaking, through incremental fine-tuning with session logs.
- Achieved a 94% classification accuracy with the primary classifier, and an overall accuracy of 89% for the entire context-aware system.

Website person in charge, Oxford University Hong Kong Scholars Association **2025 - 2026**

- Revamped the design of the association's website (https://oxhkscholars.web.ox.ac.uk) after being abandoned for over 3 years.
- Utilized the Oxford Mosaic web platform for student societies, designed a UI that is intuitive, easy to navigate, yet contains comprehensive information all members could access easily at a glance.
- Undergone a thorough review of the old website contents, integrated the old taxonomy with the new, revamped UI and taxonomy that makes the latest news and current year content the focus, while including older contents.

Events Officer, Oxford University Hong Kong Scholars Association **2025 - 2026**

- Person in charge of all social media platforms, promote events through emails, website, and social media channels
- Propose and plan events in collaboration with the president and other committee. Handle complete logistics, from initiating and managing registration processes to venue booking, scheduling, and setup
- Solely responsible for managing all communications with attendees and members regarding events
- Coordinate with vendors, suppliers, and any external partners for smooth event delivery
- Deliver regular updates and news to members through website, social media

AI Research Intern, Health Bridge International Medical Group Limited **July – Sep 2024**

- Developed an activity recognition algorithm to deploy on AI wearable health devices.
- Created a robust data pre-processing workflow handling 45+ hours of raw accelerometer and gyroscope data, implementing median filtering, z-score normalization and anomaly detection techniques to improve signal and data quality
- Implemented a CNN classifier with 2-Conv, 2-Dense layers for recognition of 7 human activities with 87% accuracy

Engineering Intern(Remote), Health Bridge International Medical Group Limited **Oct 2023 – Jun 2024**

- Engineered a RNN model for blood glucose level prediction from user-inputted daily blood glucose data with physiological data (heart rate, physical activity)
- Optimized the model to extract physiological data from Apple HealthKit API

- Provide confidence intervals for estimated future glucose levels, and push notifications of potential hyperglycemia events
- The high glucose alert achieved an F1 Score of 0.80 from an independent testing dataset
- Utilized transfer learning to enhance model personalization and accuracy with limited user-specific data.

Engineering Intern, Health Bridge International Medical Group Limited **Jun – Sep 2023**

- Designed a smart online diagnosis redirection algorithm.
- Fine-tuned an NLP model (BERT base uncased) to categorize incoming patient requests.
- Re-directed patient requests to the best available medical expert.
- Reduced average waiting time by over 20%

Co-organizer, Google Developer Groups (GDG) Devfest Oxford **2024**

- Co-organized the GDG Devfest with Gregory McGann at Exeter College, Oxford
- Coordinated venue logistics and secured appropriate facilities accommodating 100+ technology professionals and students
- Supported day-of-event operations including registration, speaker coordination, and attendee engagement

Student Ambassador, Department of Engineering Science, University of Oxford **2024 - 2026**

- Represents the Department of Engineering Science of Oxford University at various outreach events throughout the year.
- Leads facility tours that emphasize Oxford's cutting-edge research environment to prospective engineering students.

Teaching Assistant, Chinese University of Hong Kong **2023**

Assisted the lecturer in administrative work and provided guidance to students on their in-class assignments and exercises, in the "Introduction to Number Theory" and "Solving Algebraic Equations" courses

Hong Kong University of Science and Technology, HKAGE: **Jan-Feb 2020 & 2021**
Physics Enhancement Programme for Gifted Students Phase II

Ranked 16(in 2020) and 21(in 2019) in Hong Kong in the intensive Olympiad Physics training and selection programme that selects candidates representing Team Hong Kong in APhO and IPhO competitions

COURSEWORK & PRACTICALS

AI & Machine Learning in Python, Coursework Project, University of Oxford **2024**

- Supervised by Dr Yangchen Pan, Dr Jindong Gu, Dr Zhiyao Luo, Dr Kevin Sun
- Trained a regression algorithm to model housing prices in California.
- Trained a classification algorithm using SVMs, achieved over 90% validation accuracy on a dataset identifying breast cancer patient
- Trained neural networks for computer vision using PyTorch, achieved over 85% classification accuracy with the MNIST dataset

High Performance Computing, Coursework Project, University of Oxford **2024**

- Supervised by Prof Wes Armour
- Wrote algorithms in C and CUDA programming language to perform scientific calculations
- Programmed multi-core architectures using OpenMP
- Programmed many-core architectures (NVIDIA GPUs) using CUDA
- Performed the whole project in a Linux environment
- Developed on cloud supercomputers with V100 GPUs from University of Oxford

Mechanical CAD, Engineering Coursework Project, University of Oxford **2024**

- Supervised by Ms Belinda Hughes
- Designed a walking mechanism from scratch with SolidWorks
- Manufactured and assembled designed parts through 3D printing and CNC
- Passed all terrains within 30s

Instrumentation & Control Practical, University of Oxford **2024**

- Supervised by Dr. Jack Umenberger and Junaid Memon.
- Designed a PID controller for a Quanser Aero (dual-rotor two degree-of-freedom aerospace system).
- Performed modelling and simulation of the system through Simulink.
- Tuned the PID controller to control power input to the two rotors to achieve disturbance rejection and reference tracking, achieved all the design goals of zero-steady state error and a fast, smooth and stable dynamic response.

Communications Practical, University of Oxford **2024**

- Supervised by Dr. Eleanor O'Hara, Prof Dominic O'Brien
- Performed Fast Fourier Transform (FFT) algorithm with an oscilloscope to investigate incoming signals and properties of signal propagation along a coaxial cable

Electrical Practical, University of Oxford **2023**

- Supervised by Dr. Eleanor O'Hara
- Designed and built a programmable electronic music box with switch controls
- Loaded a piece of C code to the integrated circuit board to achieve programmed playback control

Mechanical Practical, University of Oxford **2023**

- Supervised by Mr R Scott
- Designed a pin-jointed truss structure withstanding 150kN of load
- Optimized the structure's cost and strength through detailed stress analysis via frame2d on MATLAB

Scholarships and Awards

Williams College Winter Study Programme Scholarship **Jan 2025**

- Jointly funded by Exeter College, Oxford & Williams College, MA, USA
- Awarded to 10 students at Exeter College, Oxford for a 10-day fully funded on-site study program at Williams College, MA, USA

AP Scholar with Distinction (College Board of the United States) **2021**

Awarded by College Board for score of 5(Top Score) in all six subjects attended

Course Scholarship, Chinese University of Hong Kong **2021**

Awarded for excellent performance in "Towards Differential Geometry" course by the Department of Mathematics, Chinese University of Hong Kong

Technical Experience

Programming Languages: Python, C++, MATLAB
ML Libraries: PyTorch, Tensorflow, Keras, Hugging Face Transformers, Pandas, Scikit-learn, Numpy, MatPlotLib
Other software: Git, SLURM, CUDA, Linux CLI, Cloud Computing via SSH

Areas of Interests

Agentic AI | Multiagent Systems | AI Safety | AI Risk Mitigation Strategies | Large Language Models (LLMs) and its applications | Natural Language Processing (NLP) | Reinforcement Learning | Multimodal AI | AI Alignment | AI Interpretability | AI Robustness

Volunteering

Committee, Access Abroad Hong Kong (AAHK) **2022 – 2025**

Working with a team of current Hong Kong students from top UK universities to provide free guidance to underprivileged students applying to top UK universities, aiming to unleash the potential of talented students who might be disadvantaged in their application due to their economic background

Academic Mentor, Access Abroad Hong Kong (AAHK) **2022 – 2023**

Provided mentorship to a student applying for Physics & Philosophy at Oxford University.

Language Proficiency

Proficient in English, Cantonese Chinese(native) and Mandarin Chinese(native)